Vanshika Uttarwar

AI Engineer | LLM Systems | AI Agents | FastAPI | AWS

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AI Engineer with 5+ years of experience building production-grade AI applications, intelligent automation systems, and scalable backend services. Experienced in LLM orchestration, AI agents, multimodal AI, speech AI, prompt engineering, FastAPI, AWS, and LangGraph, leveraging Gemini, Claude, and GPT-4o to deliver measurable business impact.

EXPERIENCE

ScribeEMR Systems Pvt. Ltd. Aug. 2025 – Present
AI Engineer Bengaluru, India

IM Nexus – AI-Powered Medical Prompt & Schema Automation Engine

- **Architected an end-to-end AI workflow** that ingests Provider Preference Guides (PDF/DOCX) and auto-generates EMR-compliant extraction prompts and JSON schemas, **reducing manual prompt engineering effort by 80%+**
- Built a **dual-LLM orchestration layer** using Gemini and Claude with intelligent fallback, retry logic, and concurrency controls, ensuring reliable generation under API rate limits and high user load
- Developed 100+ production prompts converting unstructured provider guidelines into deterministic EMR-compliant schemas.
- Created a **surgical prompt refinement engine** enabling natural-language modifications to generated schemas while enforcing anti-hallucination guardrails and preserving schema integrity
- Implemented RBAC, PostgreSQL-backed storage, and real-time status tracking for AI generation workflows.
- Collaborated with healthcare providers, operations teams, and business stakeholders to gather requirements, validate AI-generated workflows, and drive iterative product enhancements
- Created GitLab CI/CD pipelines for IM Nexus, automating build validation, testing, and release workflows for FastAPI services, improving release consistency and reducing manual effort.

WoundCare AI – Medical Scribe Pipeline

- Designed a production-grade speech-to-text pipeline using **Whisper and FastAPI**, converting provider dictation into structured medical documentation
- Developed high-precision prompt workflows extracting **25+ clinical attributes with 98%+ accuracy** while reducing hallucinations through structured validation
- Architected a **serverless AWS solution** using Lambda, Step Functions, and S3 with Pydantic-based validation for healthcare compliance
- Implemented clinical reasoning logic supporting E/M justification, debridement detection, and medical shorthand expansion

WoundCare AI – Vision Language Model Platform

- Developed a **hybrid document intelligence solution** using Gemini and GPT-4o for extracting structured data from handwritten notes and clinical PDFs
- Achieved **93–96% extraction accuracy** using ROI-based vision processing and high-resolution document rendering
- **Reduced API costs by 52%** through intelligent document chunking and optimized batch-processing workflows
- Built asynchronous FastAPI services with secure S3-backed document storage and auditability

Weekly Task Newsletter – Agentic Workflow Automation

- Built a **LangGraph-powered AI workflow** generating executive-level HRMS summaries and weekly reports. Developed an SMTP-based newsletter delivery system with scheduled weekly distribution using cron jobs
- Extended the platform with an **MCP** server exposing newsletter generation capabilities and integrated it with Antigravity for agent-driven execution and tool orchestration
- **Reduced backend response latency by 95%** through asynchronous processing, caching strategies, and API optimization
- Implemented secure tokenized access workflows for personalized newsletter delivery

Additional AI Engineering Contributions

- Developed and maintained provider-specific prompts and schema for **50+ healthcare providers**, achieving **95%+ extraction accuracy** while reducing hallucinations and ensuring compliance with individual documentation preferences.

- Built AI-powered automation workflows using **N8N** for newsletter generation, WhatsApp assistants, and operational process automation, reducing manual effort and accelerating content delivery
- Contributed to a proof-of-concept evaluating **QLoRA fine-tuning on Qwen-32B** for domain-specific clinical note generation, including dataset preparation, model training workflows, and performance evaluation
- Developed a speech analytics engine evaluating pace, clarity, tone variation, and pause patterns to improve downstream transcription quality and provider documentation accuracy

Schoolknot Aug. 2024 – Jun. 2025
Data Analyst Remote

- Engineered **SQL-based reporting pipelines** for a 1M+ record ERP platform, **improving reporting accuracy by 30%**
- Conducted exploratory data analysis to identify module adoption gaps, influencing FY25 product roadmap decisions
- Built executive **Tableau dashboards** tracking fee recovery, attendance, and operational KPIs, **improving billing efficiency by 10%**
- Developed an **AI-powered mock examination platform** that generated personalized performance analysis and automated feedback using LLM-based evaluation workflows
- Designed prompt-driven assessment pipelines to evaluate student responses, identify knowledge gaps, and recommend targeted learning paths

Encora Jun. 2021 – Aug. 2024
Software Developer / Data Analyst Pune

- Developed responsive web applications using **ReactJS, JavaScript, HTML, and CSS** for healthcare and ed-tech clients
- Built Python and SQL-based analytics solutions supporting large-scale reporting and operational decision-making
- Automated reporting pipelines using Pandas, **reducing manual effort by 40+ hours per month**
- Collaborated with cross-functional teams on API integration, database design, testing, and Agile software delivery
- Developed reusable UI components and implemented unit testing practices to improve application quality and maintainability

SELECTED PROJECTS

BuyWise – AI-Powered Product Intelligence Platform

- Built a **full-stack product comparison platform** using Next.js, Prisma, and SQLite to aggregate and analyze product listings across multiple e-commerce sources
- Developed AI-assisted product normalization workflows to identify equivalent products and standardize product specifications
- Integrated **sentiment analysis pipelines** leveraging online review content to generate concise purchase insights and highlight recurring customer concerns

SKILLS

AI & Generative AI : LLMs, Prompt Engineering, RAG, Agentic Workflows, Structured Outputs, Gemini, Claude, GPT-4o, Qwen, LangChain, LangGraph, MCP, Prompt Optimization, LLM Observability (Langfuse)

Backend : Python, FastAPI, Pydantic, SQLAlchemy

Cloud & Infrastructure : AWS (S3, EC2, Lambda, Step Functions, Glue), Serverless Architecture, Docker, Git

Data & Analytics : SQL, PostgreSQL, Snowflake, Pandas, Tableau, Excel

AI Specializations : Vision Language Models (VLMs), Whisper, Speech AI, OCR Pipelines, Document Intelligence

Frontend : ReactJS, Next.js, JavaScript, HTML5, CSS3, Tailwind CSS

EDUCATION

SMT. Kashibai Navale College of Engineering 2021
Bachelor of Engineering in Information Technology Pune, India

CERTIFICATIONS
Full Stack Data Science & AI – AlmaBetter | SQL (Advanced) | Tableau | Snowflake | Databases for Data Science (Coursera)